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Price ₹ 699

This book is dedicated to all students preparing for SSC examinations who are moving steadily toward their goals with sincerity, patience, consistency, and a calm mind. Your continuous hard work and self-discipline are strengthening the foundation of your future every day. We hope that this book will support you at every stage of your preparation, strengthen your discipline, clarity of thought, and self-confidence, and continue to inspire you to move forward until you achieve your goal.

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PREFACE

SSC MATHS KING MAKER

1

Objective of the Book



This book is a comprehensive guide prepared for various competitive examinations of SSC. In this, great emphasis has been laid on **conceptual, systematic, exam-oriented** and **result-oriented** preparation of Mathematics. Important topics like Concepts, Theory, Important Formulae, Short Tricks and Best Approaches are included so that basic to advanced level theory can be learnt and **revised** through multicolour format.

2

LATEST PATTERN PYQs



After concepts, latest PYQs have been organized in **Chapter-wise, Type-wise** and **Level-wise** format so that basic to advanced level question types can be understood easily.

3

IMPORTANT PYQs FOR UPCOMING EXAMS



Important PYQs and **important PYQs for upcoming exams** have been included, which are based on **latest exam trend** and **question pattern**. The concepts asked frequently in upcoming exams and their targeted practice are also given.

4

100 DAYS COMPLETE STUDY PLAN



This book contains **100 Days Complete Study Plan**, in which a complete study plan of each day is given. Following this plan daily, students can complete **chapter-wise** practice, **revision** and **test** and assess their preparation.

5

QR CODE VIDEO SOLUTION – LEARN WITH GAGAN SIR



Question-wise QR Code Video Solution has been provided in the book. By scanning the QR Code, video solution of any question can be learnt from Gagan Sir through **step-by-step** solution. In this, Answer, Question Approach, Concept, Application, Calculation Method, Short Tricks and **Exam Strategy** are explained in detail.

6

COMPLETE SSC EXAM COVERAGE



All types of concepts and questions of SSC CGL Tier-I & Tier-II, SSC CHSL Tier-I & Tier-II, SSC CPO, SSC MTS, SSC GD, SSC Selection Post and other SSC exams have been included in this book so that you can achieve **final revision** and perform well in the exam as well as at every **stage**.

7

OUR EFFORT, YOUR SUCCESS



This book has been prepared with the guidance of Gagan Pratap Sir in the belief that with his teaching **experience, exam analysis** and deep understanding of concepts, it will be extremely practical and **exam-oriented**. The Content Team, Typing Team and Designing Team have worked hard day and night to present it in the best possible way with zero errors. We hope that with this book you will adopt a strategy, regular practice and **100 days study plan** and achieve **speed, accuracy, concept clarity** and **confidence** and secure success.



Best wishes for your bright future and success in all your efforts





HOW TO USE THIS BOOK?

Read This Before You Start...



Every chapter in this book follows the same learning path. If you study in this sequence, your **concepts will become stronger** and your **score will improve** naturally.

Don't rush to solve questions. Build your concepts first, practice step-by-step, learn from your mistakes, and finally test yourself.

Success
is a series
of small
wins! ★

YOUR LEARNING JOURNEY

**STEP
1**



LEARN THE CONCEPTS FIRST

Start every chapter with the **Theory Section**.

- Read every concept carefully.
- Understand formulas, properties, tricks and shortcuts.
- Don't memorize—understand the logic behind every concept.



**Strong Concepts
= Easy Questions**

**STEP
2**



SOLVE PRACTICE QUESTIONS (LEVEL-WISE)

Now start solving the questions.

Questions are arranged from:

BASIC

MODERATE

ADVANCED

- Solve them in order without skipping levels.
- This will gradually improve your confidence, accuracy and speed.



**Practice Today,
Perform Better
Tomorrow!**

**STEP
3**



STUCK? SCAN THE QR CODE

If you're unable to solve any question—
Scan the QR Code.

Watch the detailed Hindi video solution by Gagan Sir.

- Complete Concept → Step-by-Step Logic
- Smart Tricks → Similar Question Approach



Don't just watch the solution.
Pause the video and solve the question yourself first.



**Doubt Today,
Clear Today!**

**STEP
4**



TAKE THE CHAPTER TEST

After completing all practice questions,
attempt the Chapter Test (Mixed Questions).
This test is designed exactly like the real SSC exam.



Concept
Clarity



Speed



Accuracy



Exam
Readiness

Treat every chapter test like a mini mock test.



**Test Yourself,
Know Yourself,
Improve Yourself!**

**STEP
5**



ANALYZE & IMPROVE

After every chapter,

- Mark the questions you got wrong.
- Revise those concepts.
- Reattempt the mistakes after 2–3 days.

★ This is where real improvement happens.



**Analyse. Revise.
Improve. Repeat.
That's the Key!**

★ GOLDEN RULE ★



CONCEPT
Understand
Clearly



PRACTICE
Level-wise
Questions



QR SOLUTION
(Only if Needed)
Clear Doubts



CHAPTER TEST
Mixed Questions
(Mini Mock Test)



ANALYZE
Find Mistakes
& Weak Topics



REVISE
Strengthen &
Master Topics

REPEAT THIS PROCESS FOR **EVERY CHAPTER**,

AND YOU'LL SEE A REMARKABLE IMPROVEMENT IN YOUR PREPARATION.

★ Stay Focused. Keep Practicing. You Will Achieve Your Goal!



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Complete theory of all SSC Math chapters

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		TYPE-II	Isosceles Triangle	8	608 – 608	
		TYPE-III	Equilateral Triangle	10	609 – 609	
		TYPE-IV	Right-angled Triangle	6	609 – 609	
		TYPE-V	Regular Hexagon	4	609 – 610	
		TYPE-VI	Quadrilatera	4	610 – 610	
		TYPE-VII	Parallelogram	4	610 – 610	
		TYPE-VIII	Rectangle	36	610 – 612	
		TYPE-IX	Square	10	612 – 612	
		TYPE-X	Rhombus	19	613 – 613	
		TYPE-XI	Trapezium	6	613 – 614	
		TYPE-XII	Circle	64	614 – 617	
		TYPE-XIII	Semicircle	4	617 – 617	
			Answer Key	–	618 – 618	
			Solution with detailed explanation	–	618 – 631	
			CHAPTER MASTER TEST	30	632 – 636	
27.	MENSURATION 3D	TYPE-I	Cube	7	637 – 637	(91-95) Days
		TYPE-II	Cuboid	26	637 – 638	
		TYPE-III	Cylinder	32	639 – 640	
		TYPE-IV	Hollow Cylinder	5	640 – 641	
		TYPE-V	Cone	41	641 – 643	
		TYPE-VI	Sphere	26	643 – 644	
		TYPE-VII	Hollow Sphere	8	645 – 645	
		TYPE-VIII	Hemisphere	14	645 – 646	
		TYPE-IX	Hollow Hemisphere	14	646 – 647	
		TYPE-X	Melting	29	647 – 648	
		TYPE-XI	Three-dimensional combination	18	648 – 649	
		TYPE-XII	Frustum	2	650 – 650	
		TYPE-XIII	Prism	29	650 – 651	
		TYPE-XIV	Pyramid	25	652 – 653	
			Answer Key	–	654 – 654	
			Solution with detailed explanation	–	654 – 669	
			CHAPTER MASTER TEST	50	670 – 676	

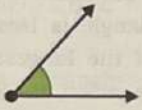
28.	STATISTICS	TYPE-I	Range	3	677 – 677	(96-97) Days
		TYPE-II	Mean	9	677 – 677	
		TYPE-III	Median	17	677 – 678	
		TYPE-IV	Mode	13	678 – 679	
		TYPE-V	Empirical	5	679 – 679	
		TYPE-VI	Variance and Standard Deviation	24	679 – 681	
		TYPE-VII	Karl Pearson's Coefficient of Skewness	5	681 – 681	
		TYPE-VIII	Quartile Deviation	3	681 – 681	
			Answer Key	–	681 – 681	
			Solution with detailed explanation	–	682 – 686	
			CHAPTER MASTER TEST	38	687 – 691	
29.	PERMUTATION AND COMBINATION	TYPE-I	Permutation	24	692 – 693	98 Days
		TYPE-II	Combination	6	693 – 693	
			Answer Key	–	693 – 693	
			Solution with detailed explanation	–	693 – 694	
30.	PROBABILITY	TYPE-I	Basic Questions	6	695 – 695	99 Days
		TYPE-II	Dice-based	8	695 – 696	
		TYPE-III	Based on coins	13	696 – 696	
		TYPE-IV	Based on cards	14	696 – 697	
		TYPE-V	Based on balls or marbles	7	697 – 697	
		TYPE-VI	Based on Set Theory	11	697 – 698	
		TYPE-VII	Miscellaneous	5	698 – 698	
			Answer Key	–	699 – 699	
			Solution with detailed explanation	–	699 – 701	
			CHAPTER MASTER TEST	10	702 – 703	

Lines & Angles

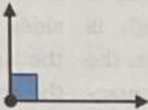
Angle: The inclination between two lines is called an angle. A figure formed by two rays or lines that share a common endpoint is called an angle. The two rays are known as the arms of the angle, and the common endpoint is called the vertex.

Types of angles

1
Acute angle



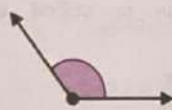
2
Right angle



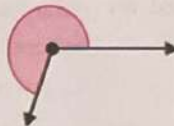
3
Straight angle



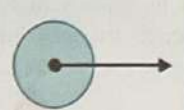
4
Obtuse angle



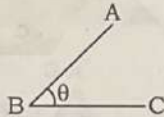
5
Reflex angle



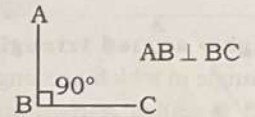
6
Complete angle



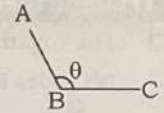
Acute angle: If an angle is greater than 0° and less than 90°
 $0^\circ < \theta < 90^\circ$ ($\angle ABC$ is an acute angle)



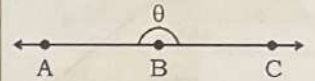
Right angle: If an angle is equal to 90° , it is called a right angle.
 $\theta = 90^\circ$ ($\angle ABC$ is a right angle.)



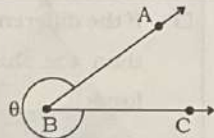
Obtuse angle: If an angle is greater than 90° and less than 180° , it is called an obtuse angle.
 $90^\circ < \theta < 180^\circ$ ($\angle ABC$ is an obtuse angle.)



Straight angle: If an angle is equal to 180° , it is called a straight angle.
 $\theta = 180^\circ$ ($\angle ABC$ is a straight angle)



Reflex angle: If an angle is greater than 180° and less than 360° , it is called a reflex angle.
 $180^\circ < \theta < 360^\circ$ ($\angle ABC$ is a reflex angle)



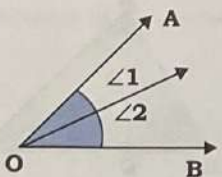
Complete angle: If an angle is equal to 360° , it is called a complete angle.



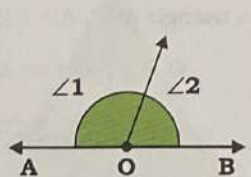
Pair angle

Pair Angles

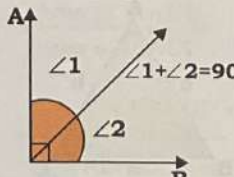
Adjacent Angle



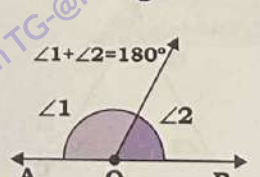
Linear Pair Angle



Complementary Angle



Supplementary Angle



Triangle

Types of Triangles

Based on Angles

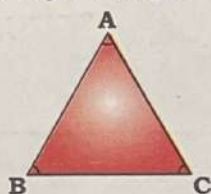
- ❖ Acute-angled Triangle
- ❖ Right-angled Triangle
- ❖ Obtuse-angled Triangle

Based on Sides

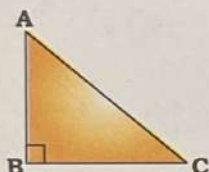
- ❖ Scalene Triangle
- ❖ Isosceles Triangle
- ❖ Equilateral Triangle

Triangle based on angle

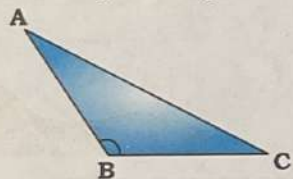
Acute-angled triangle: A triangle in which each angle is less than 90° is called an acute-angled triangle.



Right-angled triangle: A triangle in which one angle is 90° is called a right-angled triangle.

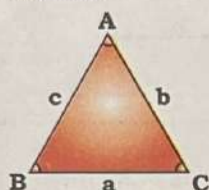


Obtuse-angled triangle: A triangle in which one angle is greater than 90° is called an obtuse-angled triangle.

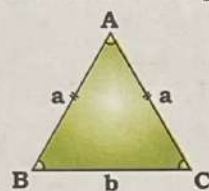


Triangle based on side

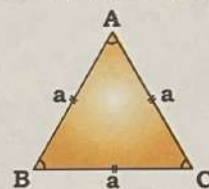
Scalene triangle: A triangle in which all three sides are of different lengths is called a scalene triangle.



Isosceles triangle: A triangle whose two sides are equal is called an isosceles triangle.



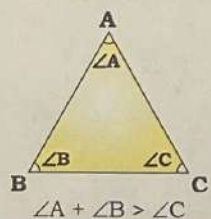
Equilateral triangle: A triangle in which all sides are equal is called an equilateral triangle.



Identifying triangles based on angles and side ratios

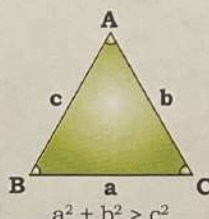
Based on angle ratio

If the sum of the ratios of two angles of a triangle is greater than the third angle, the triangle is called an acute-angled triangle.

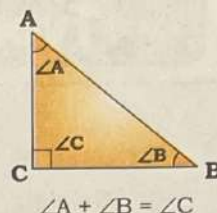


Based on arm ratio

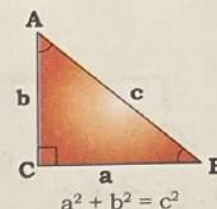
If the sum of the squares of two sides of a triangle is greater than the square of the longest side, the triangle is called an acute-angled triangle.



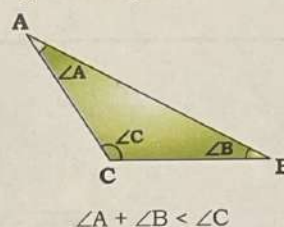
If the sum of the ratios of two angles is equal to the third angle, then the triangle is a right-angled triangle.



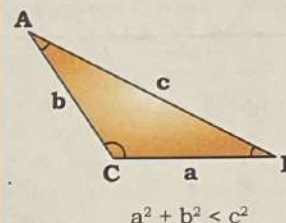
If the sum of the squares of two sides of a triangle is equal to the square of the longest side, the triangle is called a right-angled triangle.



If the sum of the ratios of two angles of a triangle is less than the third angle, the triangle is called an obtuse-angled triangle.



If the sum of the squares of two sides of a triangle is less than the square of the longest side, the triangle is called an obtuse-angled triangle.



Triangle inequality

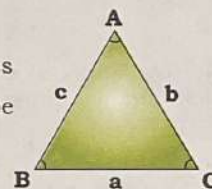
- ❑ The construction of a triangle is possible only if the sum of two sides is greater than the third side.

$$a + b > c, b + c > a, c + a > b$$

- ❑ If the difference between two sides is less than the third side, a triangle will be formed.

$$a - b < c, b - c < a, c - a < b$$

$$\therefore |c - a| < b < (c + a)$$



- ❑ The length of the longest side of a triangle is less than half of its perimeter.

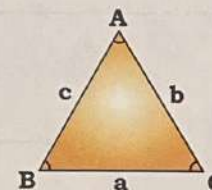
$$c < (a + b)$$

$$(c + c) < (a + b + c)$$

$$2c < \text{perimeter}$$

$$c < \frac{\text{Perimeter}}{2}$$

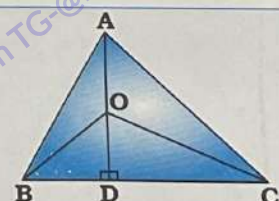
Where c = the length of the longer side.



In a triangle ABC, $AD \perp BC$,

O is a point on AD, then

$$AB^2 + OC^2 = AC^2 + OB^2$$



Based on the angle bisector of a triangle

- In triangle ABC, AE is the angle bisector of $\angle BAC$ and AD is the perpendicular drawn to side BC.

Then

$$\angle DAE = \frac{\angle B - \angle C}{2}$$

- In triangle ABC, the angle bisector of interior angle $\angle B$ and the angle bisector of exterior angle $\angle C$ meet at E.

then, $\angle BEC = \frac{1}{2} \angle BAC$

- In triangle ABC, the bisectors of $\angle B$ and $\angle C$ meet at I.

then, $\angle BIC = 90^\circ + \frac{\angle A}{2}$

- In triangle ABC, the bisectors of the exterior angles at $\angle B$ and $\angle C$ meet at O.

Then,

$$\angle BOC = 90^\circ - \frac{\angle A}{2}$$

$$\angle A = 2(90^\circ - \angle BOC)$$

- The line connecting the points formed by the intersection of two internal angle bisectors and two external angle bisectors of a triangle is always a straight line.

❖ AIO is a straight line and divides A into two equal parts.

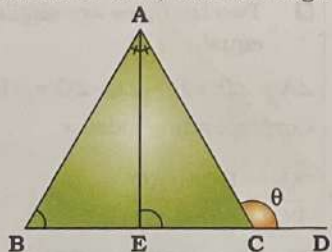
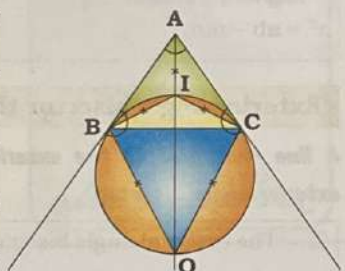
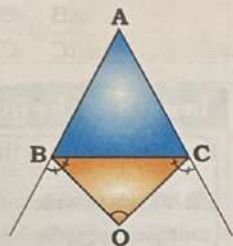
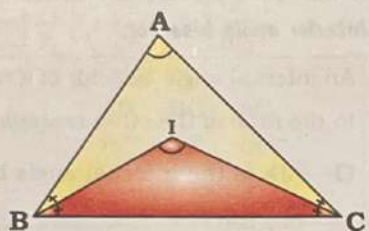
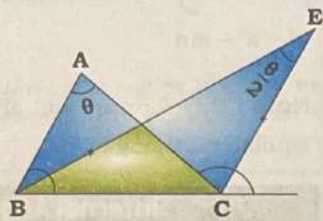
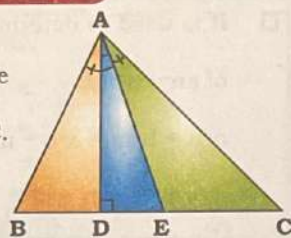
$$\angle BIC + \angle BOC = 180^\circ$$

❖ BICO is a cyclic quadrilateral.

- In a triangle ABC, side BC is extended to D, and the angle bisector of $\angle A$ meets BC at E.

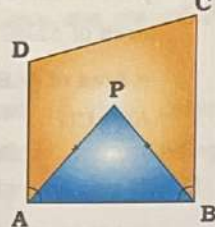
Then

$$\angle ABC + \angle ACD = 2 \angle AEC$$



- In any quadrilateral, the angle formed by the bisectors of two adjacent angles is equal to half the sum of the other two angles.

$$\angle APB = \frac{\angle D + \angle C}{2}$$



- In isosceles triangle

❖ $AB = AC$

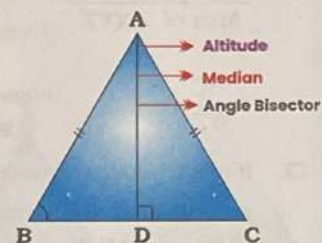
❖ $\triangle ADB \cong \triangle ADC$

❖ If $AD \perp BC \Rightarrow BD = DC$

❖ $AD \Rightarrow$ Altitude to BC,

$\Rightarrow$ Median to BC

$\Rightarrow$ Angle bisector of $\angle A$



Property of area of a triangle

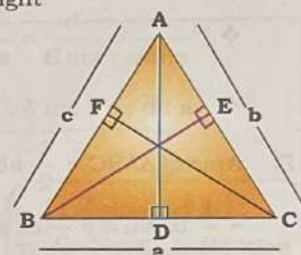
When the height of the triangle is given

- Area of $\triangle ABC = \frac{1}{2} \times \text{Base} \times \text{Height}$

- If we draw perpendiculars to all the sides of a triangle.

$$\text{Area of } \triangle ABC = \frac{1}{2} \times a \times h_1 = \frac{1}{2} \times$$

$$b \times h_2 = \frac{1}{2} \times c \times h_3$$



When height is not given in a triangle

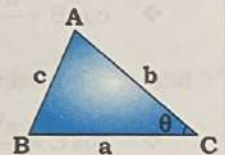
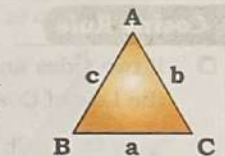
- Area of $\triangle ABC = \sqrt{s(s-a)(s-b)(s-c)}$

Where, semi-perimeter $(s) = \frac{a+b+c}{2}$

- When two sides of a triangle are known and the angle between them is given.

$$\text{Area of } \triangle ABC = \frac{1}{2} ab \sin C$$

$$= \frac{1}{2} bc \sin A = \frac{1}{2} ca \sin B$$



- When a line is drawn from a vertex of a triangle to the opposite side, the ratio in which it divides the side is the same as the ratio in which it divides the area.

$$\frac{\text{Area of } \triangle ADB}{\text{Area of } \triangle ADC}$$

$$= \frac{\frac{1}{2} \times AD \times BD \times \sin \theta}{\frac{1}{2} \times AD \times DC \times \sin(180^\circ - \theta)} = \left(\frac{BD}{CD} \right)$$

